09/21/2026

integrated inertial-magnetophoretic device? later

1 mm = 1000 μm

particle confinement ration

confinement ratio → $\lambda = \frac{a}{D_h}$ → particle diameter

→ Hydraulic diameter (Height)

"how tightly a particle is confined relative to the scale of the channel"

↳ maximum: $\lambda = 0.07$ (loosest)
↳ any smaller and inertial lift forces to weak

↳ optimal: $0.1 \leq \lambda \leq 0.3$
↳ realistic without clogging

↳ $\lambda \geq 0.5$ ↳ clogging

Size	D_h range for $0.1 \leq \lambda \leq 0.3$	
500 μm	1.67 mm - 5.00 mm	overlap 1.67-3.50
350 μm	1.17 mm - 3.50 mm	
200 μm	0.67 mm - 2.00 mm	overlap 0.67-1.00
100 μm	0.33 mm - 1.00 mm	

require more loops to seperate than the larger particles

↳ 500 μm to big for microfluidics? okied by Ryan
↳ not much literature

↳ pollen? mcc? honey?
↳ different flowers compare pollen

Pollen	μm	
very small	< 20	won't work for sieves we got
small	20 - 30	
medium	30 - 50	
large	50 - 100	
very large	> 100	

device

350 - 500 μm → inner-wall outlet
100 - 200 μm → outer-wall outlet

channels start small and get larger